

OFFICIAL MINISTRY SCIENCE CONCEPTS & CONSTANTS COMPENDIUM

Authorized Ministry of Education Reference for Physics, Chemistry & Biology

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 & EG-BACCALAUREATE SCIENCE EXAMINATIONS

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized official science concepts compendium distributed inside examination halls for Physics, Chemistry, and Biology, providing complete tables of physical constants, standard reduction potentials, solubility rules, and the standard genetic code codon table.

CORE CURRICULUM HIGHLIGHTS:

- Fundamental physical constants: h , c , e , m_e , ϵ_0 , μ_0
- Electrochemical electromotive series with standard electrode reduction potentials
- Periodic table transition metals electron configuration and oxidation states
- Standard genetic code codon table (64 codons with start and stop codons)

APPROVED CHAPTER MODULES:

- Physics Concepts, Equations & Fundamental Constants	pp. 1 40 (6 core units)
- Chemistry Periodic Table, Solubility Rules & Reaction Schemes	pp. 41 78 (6 core units)
- Biology Anatomical Tables, Hormone Cascades & Genetic Code Matrix	pp. 79 112 (6 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: National Exam Concept Compendium: Science (8p - 21.7 KB)
Full Ministerial Textbook: 112 Pages - 10.2 MB (Direct on MOE Portal)
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MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The authorized official science concepts compendium distributed inside examination halls for Physics, Chemistry, and Biology, providing complete tables of physical constants, standard reduction potentials, solubility rules, and the standard genetic code codon table. Fundamental physical constants: h , c , e , m_e , ϵ_0 , μ_0 Electrochemical electromotive series with standard electrode reduction potentials Periodic table transition metals electron configuration and oxidation states Standard genetic code codon table (64 codons with start and stop codons)

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	40% Understanding & Application Calculations, proofs & models	30% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Physics Concepts, Equations & Fundamental Constants	9 wks	20 pts	33.3%	6 core instructional units covering national standards and
Chemistry Periodic Table, Solubility Rules & Reaction Schemes	9 wks	20 pts	33.3%	6 core instructional units covering national standards and
Biology Anatomical Tables, Hormone Cascades & Genetic Code Matrix	9 wks	20 pts	33.3%	6 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: PHYSICS CONCEPTS, EQUATIONS & FUNDAMENTAL CONSTANTS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Physics Concepts, Equations & Fundamental Constants.
- [+] Synthesize and apply analytical frameworks across all 6 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Physics Concepts, Equations & Fundamental Constants.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: CHEMISTRY PERIODIC TABLE, SOLUBILITY RULES & REACTION SCHEMES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Chemistry Periodic Table, Solubility Rules & Reaction Schemes.
- [+] Synthesize and apply analytical frameworks across all 6 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Chemistry Periodic Table, Solubility Rules & Reaction Schemes.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: BIOLOGY ANATOMICAL TABLES, HORMONE CASCADES & GENETIC CODE MATRIX

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Biology Anatomical Tables, Hormone Cascades & Genetic Code Matrix.
- [+] Synthesize and apply analytical frameworks across all 6 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Biology Anatomical Tables, Hormone Cascades & Genetic Code Matrix.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

OFFICIAL MINISTRY SCIENCE CONCEPTS & CONSTANTS COMPENDIUM	
Fundamental Identity:	$\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion:	$\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard:	$\text{div}(\mathbf{J}) + \frac{\partial \rho}{\partial t} = 0$
Normalized Standard Score:	$Z = \frac{(X - \mu)}{\sigma}$
PROCEDURAL ASSESSMENT & TRANSFORMATION RULES	
Linear Operator Transformation:	$T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean:	$H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary:	$s^2 = \frac{1}{(n - 1)} \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio:	$\eta = \frac{\text{Output}_{\text{actual}}}{\text{Input}_{\text{standard}}}$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

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